177916: metasiltstone, Karonie Mine

Location and sampling

WIDGIEMOOLTHA (SH 51-14), ERAYINIA (3435)
MGA Zone 51, 458085E 6566870N

Sampled on 4 May 2004

The sample was taken from a 1 m-diameter block in the mullock heap at Karonie Mine.

Tectonic unit/relations

The unit sampled is a grey, fine-grained, weakly foliated and compositionally layered quartzofeldspathic metasiltstone, interpreted to underlie mafic and ultramafic volcanic rocks of the Black Flag Group within the Murrin Domain (Jones, 2006) of the Kurnalpi Terrane, Eastern Goldfields Superterrane, Yilgarn Craton (Cassidy et al., 2006). The mafic–ultramafic package is overlain by felsic volcanic rocks with an igneous crystallization age of 2680 ± 5 Ma (GSWA 177919; Wingate and Bodorkos, 2007b).

Petrographic description

This is a fine-grained, compositionally layered rock, and the two layer types are described separately. The bulk of the rock is composed of a fine-grained quartzofeldspathic metamorphic micromosaic with a visually estimated primary mineralogy of 55% quartz, 25% plagioclase, 15% biotite, and 5% muscovite, although the proportions of quartz and plagioclase are uncertain. Most of the micromosaic grains are less than 0.1 mm in size. Some lamellae are rich in schistose biotite flakes less than 0.3 mm long, and contain unoriented prisms of brownish-green tourmaline up to 0.5 mm long. Poikiloblastic muscovite is up to 0.5 mm in size with abundant inclusions of quartz and plagioclase. Very small, thin plates of possible graphite, and very minor fine-grained carbonate minerals, are also disseminated.

Two layers are rich in K-feldspar, with a visually estimated primary mineralogy comprising 75% K-feldspar, 15–20% plagioclase, 5% quartz, and minor biotite. K-feldspar forms poikiloblastic grains up to 0.5 mm in diameter. Plagioclase is most abundant at the margins of these layers. Lamellae rich in pyrite, tourmaline, or titanite are evident, and cross-cutting quartz veins about 0.2 mm wide are also present.

The metamorphic grade is uncertain because the observed assemblage is not diagnostic, but upper greenschist facies conditions are most likely. Weak clay clouding is seen in some of the plagioclase, and may be due to weathering.

Zircon morphology

The zircons isolated from this sample are mainly subhedral to euhedral, and clear and colourless. They are up to 150 µm long, with aspect ratios up to 4:1. Concentric growth zoning is common. Cathodoluminescence images of representative zircons are shown in Figure 1.

Analytical details

This sample was analysed on 1 August 2005, using SHRIMP-A. Eight analyses of the CZ3 standard were obtained, and indicated an external spot-to-spot (reproducibility) uncertainty of 0.00% (1σ), and a $^{238}U/^{206}Pb^*$ calibration uncertainty of 0.25% (1σ). Common-Pb corrections were applied to all analyses using contemporaneous common-Pb isotopic compositions determined by the method of Stacey and Kramers (1975).

Results

Twenty-three analyses were obtained from 23 zircons. Results are listed in Table 1 and shown in a concordia diagram (Fig. 2).

Interpretation

Most analyses range from concordant to slightly discordant (Fig. 2), and can be divided into two broad groups, based on their $^{207}Pb^*/^{206}Pb^*$ ratios and their uranium contents.

Group 1 comprises 21 analyses of 21 zircons (Table 1) with low to moderate uranium contents (36–286 ppm), which yield a weighted mean $^{207}Pb^*/^{206}Pb^*$ date of 2703 ± 5 Ma (MSWD = 1.01).

Group 2 comprises a single analysis (Table 1) with a high uranium content (982 ppm), which yields a $^{207}Pb^*/^{206}Pb^*$ date of 2578 ± 11 Ma (1σ).

Group 3 comprises a single analysis (Table 1) with a moderate uranium content (323 ppm), which yields a $^{207}Pb^*/^{206}Pb^*$ date of 1192 ± 18 Ma (1σ).

The date of 2703 ± 5 Ma for the 21 analyses in Group 1 is interpreted as the maximum age of deposition

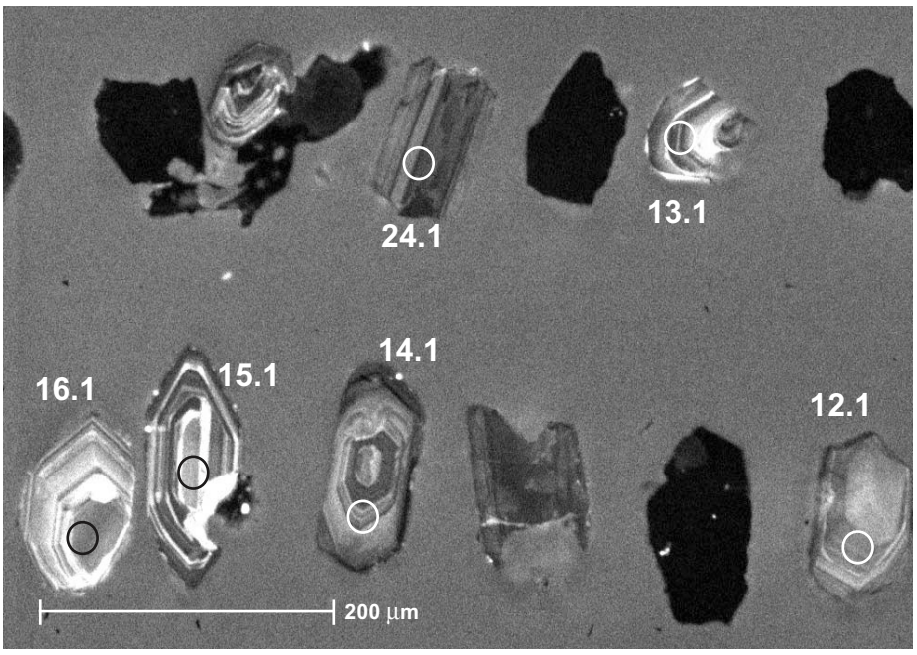

Figure 1. Cathodoluminescence image of representative zircons from sample 177916: metasiltstone, Karonie Mine. Numbered circles indicates approximate positions of analysis sites

for the siltstone protolith of this metasedimentary rock. A minimum age for its deposition is provided by the igneous crystallization age of 2680 ± 5 Ma, obtained from a felsic metavolcanic rock higher in the succession (GSWA 177919; Wingate and Bodorkos, 2007b).

The date of 2578 ± 11 Ma (1σ) obtained from the single analysis (23.1) in Group 2 is slightly discordant, and probably reflects minor ancient loss of radiogenic Pb, possibly facilitated by radiation damage to the zircon lattice as a consequence of its elevated uranium content.

The date of 1192 ± 18 Ma (1σ) obtained from the single analysis (19.1) in Group 3 is concordant, but significantly younger than expected for this rock. It is possible that this zircon underwent near-complete loss of radiogenic Pb at c. 1200 Ma, but it is considered more likely that the analysed grain is a contaminant introduced during sample preparation. The other sample on this grain mount (GSWA 177911; Wingate and Bodorkos, 2007a) also yielded a single zircon analysis significantly younger than expected for the host rock, and that is similarly likely to represent contamination.

References

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Recommended reference for this publication

WINGATE, M. T. D., and BODORKOS, S., 2007, 177916: metasiltstone, Karonie Mine; Geochronology dataset 665, *in* Compilation of geochronology data: Western Australia Geological Survey.

Data obtained: 1 August 2005
Data released: 31 May 2007

Table 1. Ion microprobe analytical results for zircons from sample 177916: metasiltstone, Karonie Mine

Grp no.	*Spot no.*	*Grain .spot*	^{238}U *(ppm)*	^{232}Th *(ppm)*	$^{232}Th/^{238}U$	*f204 (%)*	$^{238}U/^{206}Pb$ ±1σ	$^{207}Pb/^{206}Pb$ ±1σ	$^{238}U/^{206}Pb^*$ ±1σ	$^{207}Pb^*/^{206}Pb^*$ ±1σ	$^{238}U/^{206}Pb^*$ *date (Ma)* ±1σ	$^{207}Pb^*/^{206}Pb^*$ *date (Ma)* ±1σ	*Disc (%)*
1	12	12.1	86	28	0.34	0.290	1.940 ± 0.017	0.18451 ± 0.00136	1.946 ± 0.017	0.18192 ± 0.00153	2673 ± 19	2670 ± 14	–0.1
1	3	3.1	118	45	0.39	0.294	2.002 ± 0.015	0.18540 ± 0.00143	2.008 ± 0.015	0.18278 ± 0.00158	2605 ± 17	2678 ± 14	2.7
1	2	2.1	105	68	0.67	0.154	1.936 ± 0.016	0.18519 ± 0.00125	1.939 ± 0.016	0.18382 ± 0.00133	2681 ± 18	2688 ± 12	0.2
1	15	15.1	53	31	0.59	0.304	1.897 ± 0.021	0.18707 ± 0.00170	1.903 ± 0.021	0.18435 ± 0.00198	2723 ± 24	2692 ± 18	–1.1
1	5	5.1	36	15	0.44	0.339	1.939 ± 0.026	0.18765 ± 0.00211	1.945 ± 0.026	0.18463 ± 0.00259	2674 ± 29	2695 ± 23	0.8
1	11	11.1	97	35	0.37	0.112	1.913 ± 0.016	0.18563 ± 0.00128	1.91 5 ± 0.016	0.18463 ± 0.00136	2709 ± 18	2695 ± 12	–0.5
1	17	17.1	110	49	0.46	0.152	1.904 ± 0.014	0.18615 ± 0.00115	1.907 ± 0.015	0.18479 ± 0.00123	2718 ± 17	2696 ± 11	–0.8
1	7	7.1	70	34	0.50	0.565	1.937 ± 0.028	0.18983 ± 0.00155	1.948 ± 0.028	0.18479 ± 0.00190	2670 ± 31	2696 ± 17	1.0
1	6	6.1	81	47	0.60	0.286	1.866 ± 0.017	0.18745 ± 0.00140	1.872 ± 0.017	0.18490 ± 0.00159	2759 ± 20	2697 ± 14	–2.3
1	16	16.1	81	37	0.48	0.120	1.894 ± 0.048	0.18601 ± 0.00134	1.896 ± 0.048	0.18493 ± 0.00146	2731 ± 57	2698 ± 13	–1.2
1	13	13.1	50	21	0.44	0.342	2.004 ± 0.022	0.18840 ± 0.00183	2.011 ± 0.022	0.18535 ± 0.00207	2602 ± 24	2701 ± 18	3.7
1	22	22.1	173	40	0.24	0.079	1.934 ± 0.012	0.18614 ± 0.00095	1.935 ± 0.012	0.18544 ± 0.00100	2685 ± 13	2702 ± 9	0.6
1	20	20.1	83	34	0.42	0.131	1.862 ± 0.016	0.18662 ± 0.00135	1.865 ± 0.016	0.18544 ± 0.00144	2768 ± 19	2702 ± 13	–2.4
1	1	1.1	59	26	0.45	0.095	1.955 ± 0.021	0.18635 ± 0.00182	1.957 ± 0.021	0.18551 ± 0.00186	2661 ± 23	2703 ± 17	1.6
1	14	14.1	137	58	0.44	0.126	1.990 ± 0.015	0.18690 ± 0.00107	1.993 ± 0.015	0.18578 ± 0.00109	2622 ± 16	2705 ± 10	3.1
1	4	4.1	286	112	0.41	0.055	1.910 ± 0.017	0.18682 ± 0.00075	1.911 ± 0.017	0.18632 ± 0.00078	2713 ± 20	2710 ± 7	–0.1
1	10	10.1	139	130	0.97	0.198	2.029 ± 0.038	0.18843 ± 0.00125	2.033 ± 0.038	0.18666 ± 0.00137	2578 ± 39	2713 ± 12	5.0
1	21	21.1	100	44	0.45	0.328	1.874 ± 0.015	0.18988 ± 0.00126	1.880 ± 0.015	0.18695 ± 0.00144	2750 ± 18	2716 ± 13	–1.3
1	9	9.1	185	87	0.49	0.125	1.884 ± 0.011	0.18830 ± 0.00147	1.887 ± 0.011	0.18719 ± 0.00151	2742 ± 13	2718 ± 13	–0.9
1	8	8.1	151	46	0.31	0.140	1.953 ± 0.013	0.18881 ± 0.00115	1.956 ± 0.013	0.18756 ± 0.00121	2662 ± 14	2721 ± 11	2.2
1	18	18.1	36	11	0.33	0.323	1.962 ± 0.030	0.19157 ± 0.00211	1.968 ± 0.030	0.18869 ± 0.00263	2649 ± 33	2731 ± 23	3.0
2	23	23.1	982	591	0.62	0.034	2.155 ± 0.010	0.17240 ± 0.00113	2.156 ± 0.010	0.17210 ± 0.00113	2456 ± 9	2578 ± 11	4.7
3	19	19.1	323	146	0.47	0.141	4.957 ± 0.034	0.08097 ± 0.00065	4.964 ± 0.035	0.07978 ± 0.00073	1183 ± 8	1192 ± 18	0.7

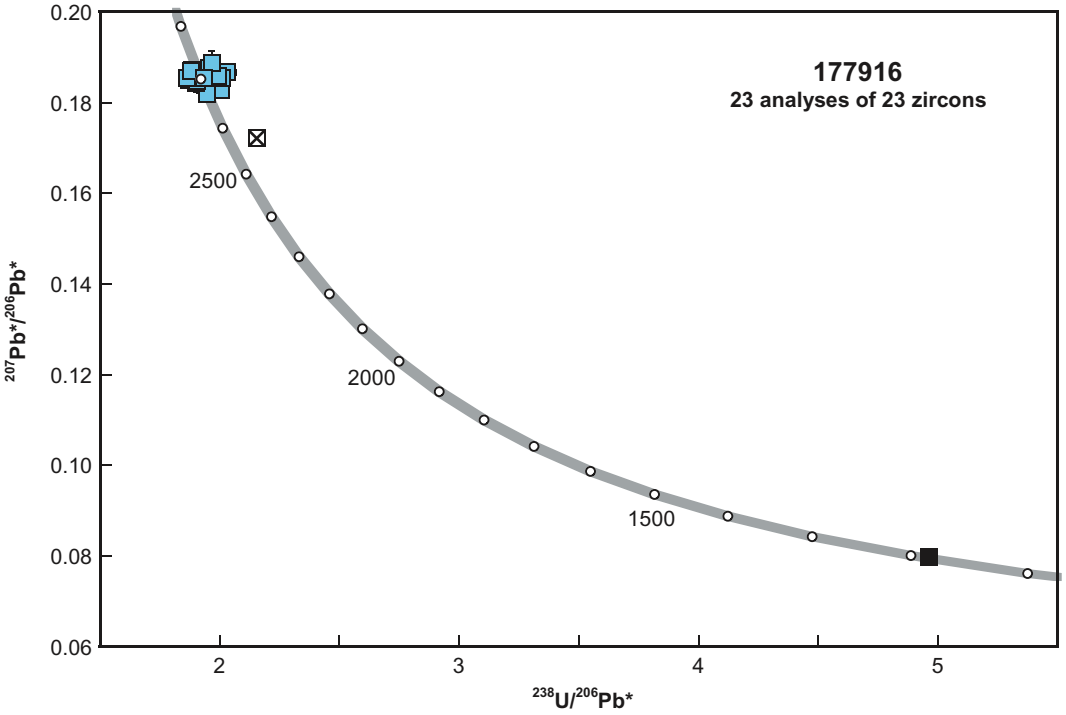

Figure 2. U–Pb analytical data for sample 177916: metasiltstone, Karonie Mine. Blue squares indicate Group (youngest detrital zircons); crossed square indicates Group 2 (radiogenic Pb loss); black square indicates Group 3 (probable contaminant)

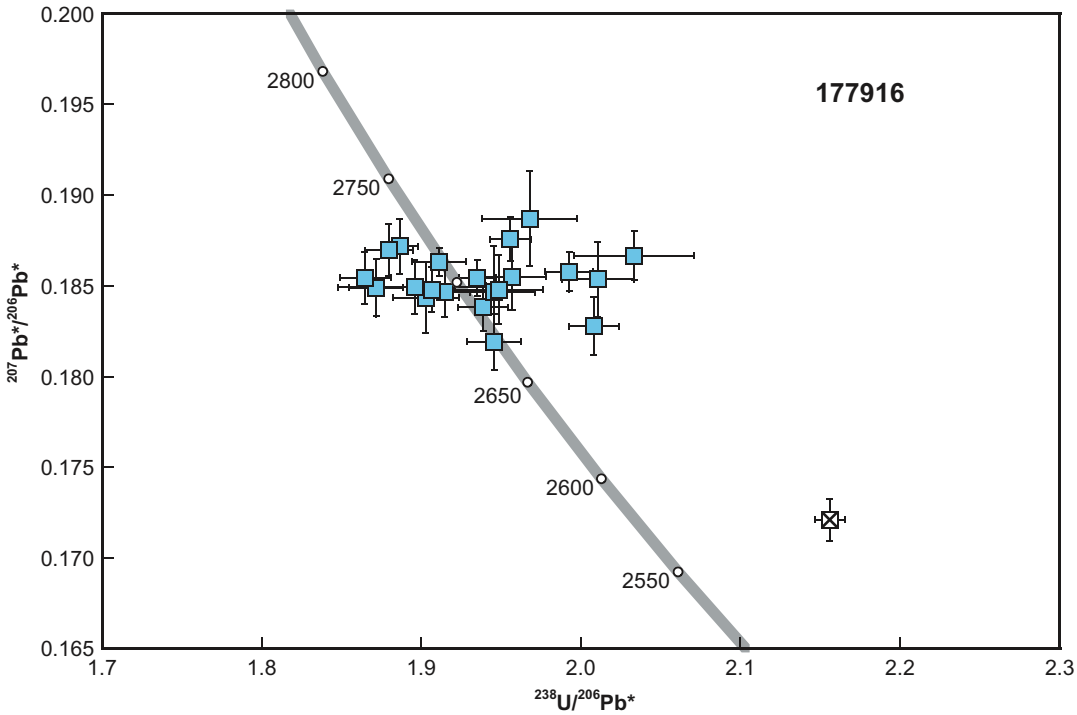

Figure 3. Expanded view of U–Pb analytical data for sample 177916: metasiltstone, Karonie Mine. Symbols as in Figure 2